Journal of King Saud University Computer and Information Sciences (2025) 37:253
https://doi.org/10.1007/s44443-025-00282-1

ORIGINAL PAPER

Attention-based federated learning for multi-dimensional personalized recommendation system

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Received: 19 June 2025 / Accepted: 10 September 2025 / Published online: 14 October 2025

Abstract

Federated learning offers a privacy-preserving solution for personalized recommendation by enabling collaborative model training across multiple clients without sharing sensitive user data. However, existing federated recommendation systems still struggle with three intertwined challenges: capturing multi-dimensional user preferences, modeling complex user–item interactions in distributed settings, and balancing the trade-off between privacy protection and recommendation accuracy. To this end, we propose Attention-based Personalized Federated Recommendation (APFR), a novel framework that integrates attention mechanisms to enhance both personalization and contextual understanding. APFR introduces a personalized score function and user embedding strategy to more precisely represent diverse user preferences, while multi-head attention is employed to capture intricate user–item relationships and contextual dependencies. Furthermore, APFR incorporates differential privacy techniques, such as Laplace and Gaussian noise, to ensure robust privacy protection with minimal accuracy loss. Experimental results on benchmark datasets demonstrate that APFR achieves superior performance in recommendation accuracy and privacy preservation, compared to state-of-the-art methods. The code and datasets are available at https://github.com/lyg2618/APFR.

Keywords Federated learning · Recommendation systems · Personalization · Attention mechanism

1 Introduction

Federated recommendation systems have emerged as a critical area at the intersection of recommendation algorithms and federated learning (Sun et al. 2024; Tan et al. 2022; Blanco-Justicia et al. 2021). Unlike traditional recommendation systems that aggregate user data on centralized servers, federated systems collaboratively train models across distributed devices (Yang et al. 2022; Javeed et al. 2023). This approach enhances user data security by keeping private information localized, addressing growing concerns over privacy breaches (Yang et al. 2020). By utilizing the com-

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putational resources of user devices, excessive reliance on a central server is avoided. It is especially useful for handling large amounts of user data, allowing the system to easily scale to handle the growth of the user base (Bonawitz 2019).

Despite advancements in federated recommendation systems, existing models often fall short of capturing users' diverse perceptions of the same items (Li et al. 2023; Zhang et al. 2023). These systems primarily focus on sharing item embeddings while keeping user embeddings private and localized. This design, while privacy-preserving, can lead to suboptimal personalization since individual user preferences are not fully exploited.

To improve recommendation quality, it is essential to better model individual differences among users. Zhang et al. (2023) proposed a PFedRec, which designed a dual personalization mechanism to achieve fine personalization at the user and item level. This approach not only utilized a specialized score function to capture user preferences by local training but also through lightweight fine-tuning of shared item embeddings. However, although PFedRec has demonstrated advantages, there are still some shortcomings. First of all, the trade-off between user privacy and recommendation accuracy is still not completely solved, which is manifested in that the embedded information of transmitted items may expose certain user behavior patterns. Secondly, the current privacy protection effect is relatively limited (Chai et al. 2020; Lin et al. 2020). Although the differential privacy method can alleviate some risks, its impact on recommendation performance still needs to be further optimized.

This paper presents an Attention-based Personalized Federated Recommendation model, namely APFR, to address the above issues. APFR integrates model personalization and attention mechanisms to address the limitations of existing approaches. APFR dropped the traditional item embedding sharing while utilizing personalized score and user embedding to improve the personalized recommendation capability in Federated Learning. Each client could better capture user preferences by using the proposed approach and avoid sensitive information leaks.

The framework leverages scaled dot-product attention and multi-head attention to improve its understanding of sequence context. These mechanisms enhance the classification performance of client-side models, enabling more accurate and nuanced recommendations. Furthermore, APFR is compatible with differential privacy algorithms, ensuring robust privacy protection with minimal impact on performance.

Through comparisons on public datasets, the proposed framework demonstrates superior recommendation accuracy and privacy preservation compared to baseline methods. By addressing critical gaps in existing federated recommendation systems, this work advances the state-of-the-art and provides practical solutions for balancing personalization and privacy in distributed settings.

Our contributions can be summarized as follows:

- We integrate attention mechanisms within a federated recommendation paradigm, enabling the model to capture complex user-item and item-item interactions across distributed clients
- We design a novel personalized score function and user embedding strategy that jointly model multi-dimensional user preferences, improving recommendation accuracy under strict privacy constraints.
- We investigate the impact of multiple differential privacy mechanisms on recommendation quality, and demonstrate that our framework can achieve a better trade-off between privacy preservation and utility than existing baselines.

The rest of this paper is organized as follows. Section 2 reviews related work. Section 3 provides preliminaries of the objective function. Section 4 presents the proposed APFR framework in detail. Section 5 reports the experimental setup and results. Section 6 concludes the paper and discusses future research directions.

2 Related works

2.1 Personalized federated learning

Federated Learning has emerged as a significant advancement in training artificial intelligence models under the paradigm of preserving data privacy, as highlighted in multiple surveys (Zhang et al. 2021; Banabilah et al. 2022). By decentralizing the data processing to local devices, Federated Learning avoids the need to share private data among participants, yet allows for collaborative model training (Banabilah et al. 2022; Aledhari et al. 2020). However, Federated Learning is not devoid of drawbacks. The decentralized nature of Federated Learning can lead to challenges such as vulnerability to data poisoning attacks and model inference attacks, which can compromise both the model's integrity and participants' data privacy (Tan et al. 2022; Lyu et al. 2020).

These inherent vulnerabilities in Federated Learning have led to the development of Personalized Federated Learning (PFL), which aims to tailor the learning process to individual participants, thereby enhancing model robustness and data privacy. PFL leverages techniques like model-agnostic meta-learning and Moreau envelopes to provide not only personalization but also theoretical guarantees on privacy and performance (Fallah et al. 2020; Dinh et al. 2020). Current research in PFL is focused on addressing the heterogeneity in participant data by personalizing the learning models to fit individual data distributions, which helps mitigate some of the critical threats identified in traditional Federated Learning systems (Fallah et al. 2020; Kulkarni et al. 2020).

Nevertheless, PFL is still facing challenges such as scalability to large numbers of users, maintaining the balance between personalization and privacy, and the computational overhead of implementing complex personalization algorithms on local devices (Kulkarni et al. 2020). These issues present ongoing research opportunities and necessitate further advancements in algorithmic efficiency and robust privacy-preserving mechanisms to fully realize the potential of PFL in practical applications.

2.2 Transformer-based recommender systems

Transformer was proposed by Vaswani (2017), replacing typical Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks with a self-attention mechanism, which greatly enhances the efficiency of processing long sequential data. This innovation allows the Transformer to process input data in parallel, significantly increasing training speed and improving performance.

Since its introduction, Transformer has been applied not only in the Natural Language Processing (NLP) domain but also in recommender systems. The self-attention mechanism helps the model understand the relationships between users' short- and long-term behaviors (Raza et al. 2024; Zhou et al. 2023). SASRec (Kang and McAuley 2018) is an early example that uses this mechanism to dynamically adjust the weights of users' historical behaviors, performing well on both sparse and dense datasets. BERT4Rec (Sun et al. 2019) further exploits the self-attention mechanism by applying bidirectional self-attention to learn users' historical interaction sequences. BERT4Rec also predicts randomly masked items in the Cloze task, utilizing context information around the masked items for better predictions. Beyond sequential recommendation, Transformer plays a crucial role in session-based recommendation (Lu et al. 2022). Transformers4Rec (Souza Pereira Moreira et al. 2021) excels in session-based recommendation in e-commerce and news domains (Moreira et al. 2021), particularly in predicting users' next clicks.

Furthermore, the Transformer can be used for explainable recommendations. Specifically, PETER (Li et al. 2021) applies user and item IDs, combining context prediction tasks to connect these IDs and generate explainable sentences. These sentences are personalized to reflect users' interests and item characteristics. Recently, transformers, with their powerful self-attention mechanisms, have also been applied to multimodal recommendation tasks. In MMGRec (Liu et al. 2024), the transformer processes multimodal item information to generate Rec-IDs and uses relation-aware self-attention to capture relationships between features, thereby improving recommendation accuracy.

3 Preliminary

3.1 Objective function

APFR treats each user as a client in Federated Learning and views the recommendation task for each user as a personalized federated recommendation problem. Concretely, the item embedding module E_i serves as a global component to learn common item information, while the score function S_i is used to support the learning of personalized decision-making logic for the i_{th} user. To further capture the differences between users and achieve preference-consistent item embeddings, the APFR has utilized a dual-layer optimization objective function (Zhang et al. 2023):

$$\begin{aligned} &\min_{\theta^m,\{\theta_i\}_{i=1}^N} \sum_{i=1}^{N} \alpha_i L_i(\theta_i; r, \hat{r}) \\ &\text{s.t.} \quad \theta_i = (\theta^m - \nabla_{\theta^m} L_i, \theta_i^s - \nabla_{\theta_i^s} L_i), \end{aligned} \tag{1}$$

where $\theta := (\theta_i^m, \theta_i^s)$ denotes the personalized parameter of E_i and S_i; α_i is the weight for the loss of the i_{th} client; L_i evaluates the local data D_i of i_{th} client; r is the user's rating and $\hat{r}$ denotes the prediction value of score function. In this framework, APFR first imports the global item embeddings E into the personalized item embedding module E_i, then learns a lightweight local score function S_i for personalized predictions.

3.2 Loss function

For the recommendation task, we denote the predicted result for the i_{th} user and j_{th} item as $\hat{r}_{ij}$, which represents the predicted score (or interaction probability) given by the model. The specific scoring function used to obtain $\hat{r}_{ij}$ will be introduced in Section 4.3.

Specifically, this paper applies a typical implicit feedback recommendation task, where if the i_{th} user interacts with the j_{th} item, $r_{ij} = 1$; otherwise, $r_{ij} = 0$. Utilizing the binary nature of implicit feedback, this paper defines the loss function for the i_{th} user as binary cross-entropy loss:

$$L_i(\theta_i; r); \hat{r} = - \sum_{(i,j) \in D_i} \log \hat{r}_{ij} - \sum_{(i,j') \in D_i^-} - \log(1 - \hat{r}_{ij}), \tag{2}$$

where D_i^- represents the set of negative instances for user i, and we use Binary Cross-Entropy (BCE) loss for a simplified description. Notably, to effectively construct D_i^-, we initially group all non-interacted items as:

$$\tau_i^- = \tau \backslash \tau_i, \tag{3}$$

where τ indicates the complete list of items, and τ_i denotes the set of items interacted with by the i_{th} user (Zhang et al. 2023). Then, based on the number of observed interactions, set the sampling ratio and uniformly sample negative instances from τ_i^- to obtain D_i^-.

4 Proposed method

4.1 Overview

This paper introduces the APFR framework, which utilizes the Transformer to learn multiple user-specific recommendation models deployed on end-user devices simultaneously. The framework of APFR is shown as Fig. 1. Firstly, the client transforms item data into item embeddings. After that, a multi-head attention mechanism is employed to capture complex relationships between items and user preferences. This mechanism generates an attention distribution, which allows the model to focus on the most relevant features of the items, ultimately improving the quality of rating prediction for each user. The loss is calculated by comparing the actual ratings with the predictions, and the weights of the item embeddings are updated based on the loss. Then, the central server aggregates the uploaded parameters from all participating clients using a weighted averaging strategy. Finally, the aggregated global model parameters are redistributed to all clients, who use them to update their local models. This iterative process continues until the model reaches satisfactory performance or a predefined stopping criterion is met.

4.2 Attention mechanism design

In Federated Learning, policies and neural networks are prone to instability, and averaging model parameters may exacerbate this instability. Inspired by the Transformer model (Ma et al. 2023), this paper employs a multi-head self-attention mechanism in the design of neural networks. This approach enhances the interrelationships among item embeddings and facilitates the integration of context.

4.2.1 Scaled dot-product

The Scaled Dot-Product Attention takes queries, keys with dimension d_k, and values with dimension d_v as input. The mechanism computes the dot product between the query and all keys, divides each result by $\sqrt{d_k}$, and applies the softmax function to obtain the weights for the values (Ma et al. 2023).

In practice, this paper computes the attention function for a set of queries, keys, and values simultaneously, taking the corresponding matrices Q, K, and V for queries, keys, and values as inputs (Ma et al. 2023). The computed output matrix is as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V \tag{4}$$

Scaled Dot-Product Attention includes a scaling factor that adjusts the attention weights to ensure numerical stability. When the value of $\sqrt{d_k}$ is large, the magnitude of the dot products increases, which can lead to a zero gradient during the backward pass of the softmax function, causing gradient vanishing. By scaling down the large $\sqrt{d_k}$ value, specifically multiplying the dot product by $\frac{1}{\sqrt{d_k}}$, the variance of the score computation changes from d_k to 1. This adjustment prevents the gradients in the softmax backward calculation from trending toward zero, thus ensuring stability during training.

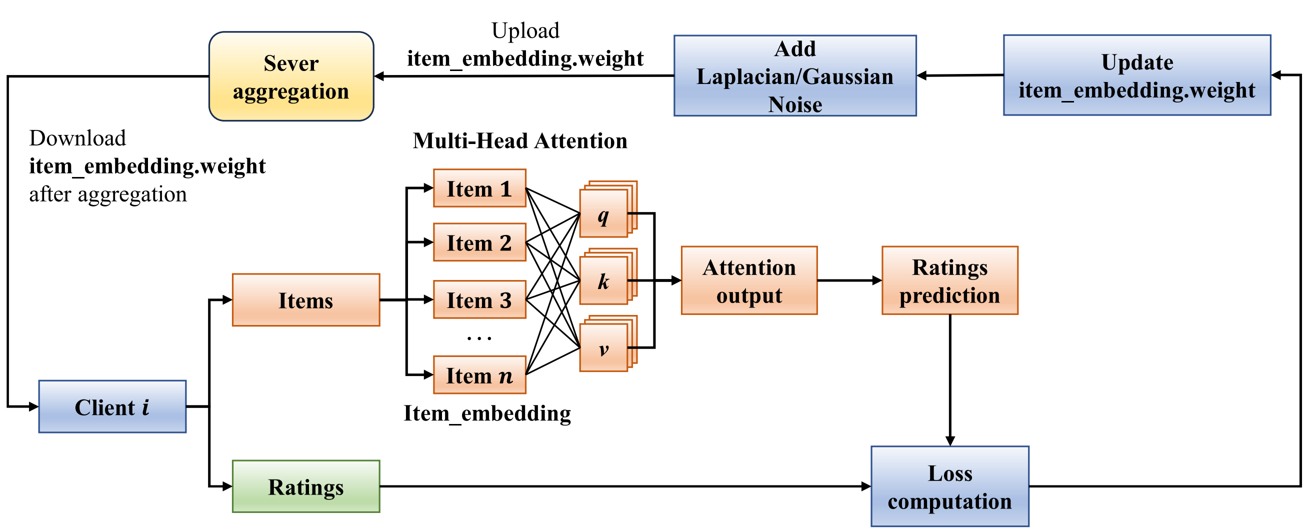


Fig. 1 Framework of Attention-based Personalized Federated Recommendation (APFR)

4.2.2 Multi-head attention mechanism

Models utilizing self-attention mechanisms often encounter a problem where the attention is excessively focused on the position itself when encoding information about the current location. To address this, this paper enhances the Scaled Dot-Product Attention by employing a multi-head attention mechanism. This approach divides the model into multiple heads, creating several subspaces, allowing the model to attend to different aspects of information. Consequently, the outputs from the attention layer include encoded representations from these diverse subspaces, enhancing the model's expressive capability (Ma et al. 2023). The definition of the multi-head attention mechanism is as follows:

$$\begin{aligned}&\text{MultiHead}(Q, K, V) = \text{Concat}(head_1, \ldots, head_h)W^O\\&\text{s.t. } head_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)\end{aligned} \tag{5}$$

where W_i^Q, W_i^K and W_i^V indicate the corresponding elementary matrix of Q, K and V. In this paper, $h = 8$ is selected as the parameter of the multi-head attention mechanism.

4.3 Score function

In the APFR framework, the score function $S_i(\cdot)$ is designed to model user-specific preference patterns effectively. Unlike conventional approaches that employ shallow neural networks or simple linear transformations, we implement S_i using a multi-head attention mechanism, enabling the model to capture richer and more complex user decision logic.

Given the item embedding e^j, the score function for user i is formulated as:

$$\hat{r}_{ij} = S_i(E_i(e^j)) = \sigma\big(\text{MultiHead}_i(Q_i, K_i, V_i)\big), \tag{6}$$

where $Q_i = E_i(e^j)W_i^Q$, $K_i = E_i(e^j)W_i^K$ and $V_i = E_i(e^j)W_i^V$ correspond to the query, key, and value projections for user i, MultiHead$(\cdot)$ follows the definition in (5), $\sigma(\cdot)$ is the sigmoid activation function.

By leveraging the multi-head attention structure, the score function can attend to multiple representation subspaces within the item embedding, allowing the model to better capture the diverse and non-linear relationships underlying user preferences. This leads to a more expressive and flexible personalization approach.

4.4 User privacy protection based on differential privacy algorithms

Considering the need for data privacy protection when clients upload data to servers, this paper integrates local differential privacy techniques into the APFR algorithm. It achieves this by implementing two types of noise functions designed for differential privacy to protect user privacy. These functions are the Laplace noise function (Dwork et al. 2014, 2006) and the Gaussian noise function (Dwork et al. 2014).

4.4.1 Laplace noise

The probability density function (PDF) of Laplace noise can be presented as:

$$\text{Laplace}(0, \lambda) = \frac{1}{2\lambda}e^{-\frac{\theta}{\lambda}}, \tag{7}$$

where λ is the scale parameter. It controls the size of Laplace noise. Laplace noise is widely applied in scenarios such as statistical queries (e.g., mean, standard deviation) and large-scale data sampling and aggregation, ensuring privacy protection while maintaining the accuracy of results (Huang et al. 2020). In addition, Laplace noise can also be utilized to provide differential privacy in deep neural networks by adaptively adding noise based on feature relevance (Phan et al. 2017).

4.4.2 Gaussian noise

Furthermore, the PDF of Gaussian noise can be written as:

$$\text{Gauss}(0, \lambda) = \frac{1}{\lambda\sqrt{2\pi}}e^{\frac{\theta^2}{2\lambda^2}}, \tag{8}$$

where λ is the variance of the distribution, controlling the scale of noise. Gaussian noise is commonly used in differential privacy to safeguard sensitive data in applications such as deep learning, where it helps to mitigate privacy leakage over multiple training epochs by controlling the cumulative privacy loss (Ouadrhiri and Abdelhadi 2022). In Federated Learning, Gaussian noise is particularly effective for protecting gradient updates while balancing the trade-off between privacy and model accuracy, making it a preferred choice for high-accuracy scenarios (Kim et al. 2021).

By adding noise, the accuracy of model training decreases; however, conversely, security performance improves. Achieving higher security with minimal loss in recommendation performance is one of the most outstanding properties of differential privacy algorithms.

4.5 Model training

To solve the optimization problem described by the objective function in this section, this paper employs an alternative optimization algorithm to train the model. As shown in Algorithm 1, once the client receives the item embeddings from the server, it first replaces these embeddings with the global embeddings and updates the score function while keeping the item embedding module fixed. Subsequently, the client updates the item embeddings based on the updated personalized score function. Finally, the updated item embeddings are uploaded to the server for global aggregation.

Algorithm 1 Server execute and client update.

function ServerExecute
  Initialize item embedding θ^m and score function θ^s
  for $t = 1, 2, \ldots$ **do**
    $S_t \leftarrow$ Randomly select a subset of clients, $S_t \subseteq \{1, 2, \ldots, N\}$, with $|S_t| = n$;
    for client $i \in S_t$ **in parallel do**
      $\theta_i^m \leftarrow \text{ClientUpdate}(i, \theta_m)$ ▷ Distribute global item embedding to client for update
      $\theta^m \leftarrow \frac{1}{n}\sum_{i=1}^{n}\theta_i^m$ ▷ Global aggregation over n local updated item embeddings
    end for
  end for
end function
function ClientUpdate(i, θ^m)
  Initialize θ_i^m with θ^m
  Initialize θ_i^s with the latest update
  Count all uninteracted items set τ_i^- with (3)
  Sample negative instances set D_i^- from τ_i^-
  $B \leftarrow$ (split $D_i \cup D_i^-$ into batches of size B)
  for $e = 1$ **to** E **do**
    for batch $b \in B$ **do**
      Initialize Q, K, V
      Attention output = MultiHead(Q, K, V) with (5)
      r = Sigmoid(Attention output)
      Compute $L_i(\theta_i; r, \hat{r})$ with (2) ▷ Model loss of batch data b
      $\theta_i^s \leftarrow \theta_i^s - \eta\nabla_{\theta_i^s} L_i$ ▷ Score function update
      Compute $L_i(\theta_i; r, \hat{r})$ (2) ▷ Model loss with the updated θ_i^s
      $\theta_i^m \leftarrow \theta_i^m - \eta\nabla_{\theta_i^m} L_i$ ▷ Post-tuning for personalized item embedding module
      $\theta_i^m \leftarrow \theta_i^m$ – Laplace$(0, \lambda)$ or Gauss$(0, \lambda)$
    end for
  end for
  return θ_i^m to server
end function

At the start of federated optimization, the server initializes the model parameters, which serve as the initial parameters for all client models. In each round, the server randomly selects a group of clients and assigns the global item embeddings θ_m to them. When the local training on the clients is complete, the server collects the updated item embeddings from each client to perform global aggregation. Algorithm 1 summarizes the entire process, and the workflow is illustrated in Figure.

In the basic recommendation scenario, where only a user-item interaction matrix exists, the item embedding module E_i often dominates the parameter volume in the recommendation model due to the large scale of the item set. This poses challenges for devices with limited computational resources. For such devices, it may be beneficial to compress and optimize the recommendation model. Techniques such as model pruning, quantization, or distillation could be employed to reduce the model's parameter count and enhance computational efficiency. However, the set of items each user interacts with in the dataset is typically much smaller than the complete item set. Based on this observation, this paper suggests that each device only needs to maintain a set of positively interacted items and perform sampling of the negative class samples using (6), instead of storing the complete item embedding module. By maintaining only the positively interacted item set and sampling the negative sample set on each device, it is possible to significantly reduce the number of parameters that need to be stored and updated on the device, thereby improving the operational efficiency of the recommendation model on endpoint devices.

5 Experiments

5.1 Experiment setup

5.1.1 Datasets

We conducted experiments on four datasets: MoviesLens-100K (Harper and Konstan 2015), MoviesLens-1M (Harper and Konstan 2015), Lastfm-2K (Cantador et al. 2011) and Amazon Videos (Ni et al. 2019). Specifically, the MovieLens website collected two datasets with different sizes, each user in the dataset has at least 20 ratings. Lastfm-2K is a music recommendation dataset. Amazon Video contains product comments and metadata. We exclude the users in Lastfm-2K and Amazon Video who have less than 5 interactions. The detailed information of datasets is shown as Table 1. We followed the leave-one-out cross-validation principle in data split (He et al. 2017).

5.1.2 Baselines

We compare the performance of APFR with the following centralized and federated recommendation models.

MF (Koren et al. 2009) is a typical centralized recommendation model, which factorizes user–item interaction matrix

Table 1 Detail information of datasets

Dataset	Interactions	Users	Items	Sparsity
MovieLens-100K	100,000	943	1,682	93.70%
MovieLens-1M	1,000,209	6,040	3,706	95.53%
Lastfm-2K	185,650	1,600	12,454	99.07%
Amazon-Video	63,836	8,072	11,830	99.93%

into two low-dimensional embedding matrices to represent potential features of users and items. Inner product is utilized for predicting the user's rating of an item.

FedMF (Chai et al. 2020) is an extension of MF in the federated learning framework. It keeps the user embedding vector in local machines, while the server is responsible for handling global item embeddings. This approach is suitable for recommendation tasks that need privacy protection.

FedNCF (Perifanis and Efraimidis 2022) is a federal variant of Neural Collaborative Filtering (NCF). It captures the non-linear relationships between users and items through neural networks. The embeddings of the user and item are trained in local machines to avoid privacy disclosure.

FedPerGNN (Wu et al. 2022) combines Graph Neural Networks (GNNs). It improves recommendation quality by modeling graph structure features of the user–item relations. A personalized module is introduced which allows each client to adjust model parameters according to their data.

PFedRec (Zhang et al. 2023) generates personalized user models by combining the global model with users' local data. This approach tries to solve the problem of heterogeneous user preference by fine-tuning the parameters of the global model.

5.1.3 Evaluation metrics

We utilized HR@10 (Hit Ratio) and NDCG@10 (Normalized Discounted Cumulative Gain) in our experiments to evaluate the model performance. HR@10 evaluates whether the recommended list contains the target item that the user is actually interested in:

$$\mathrm{HR@10} = \frac{\sum_{u=1}^{m} \mathrm{hit}_u}{\sum_{u=1}^{m} N_u}, \tag{9}$$

where hit_u denotes the number of hits in the Top-10 recommended list of user u; N_u is the actual number of recommended target items. In addition, NDCG@10 assesses the ranking performance of the recommended list:

$$\mathrm{NDCG@10} = \frac{1}{\mathrm{IDCG@10}} \sum_{i=1}^{10} \frac{2^{rel_i} - 1}{\log_2(i+1)}, \tag{10}$$

where rel_i represents the relevance score of $i - th$ item in the list; IDCG@10 refers to the Ideal Discounted Cumulative Gain.

5.1.4 Implementation details

In the training set, the negative sampling number is 4, which means 1 positive sample will be trained with 4 negative samples. In terms of the validation set and testing set, the negative sampling number is 99. The experiment results in performance comparison were averaged from 5-time executions. Detailed parameters setting is shown as Table 2.

5.2 Performance comparison

Table 3 presents the performance of APFR and five baseline methods (MF, FedMF, FedNCF, FedPerGNN, and PFedRec) across four benchmark datasets, evaluated by both HR and NDCG metrics. First, APFR consistently achieves the best or near-best performance across all datasets and both evaluation metrics. Specifically, APFR obtains the highest HR scores on MovieLens-100K (74.12%), MovieLens-1M (73.72%), Lastfm-2K (84.12%), and Amazon Video (60.55%). The improvements over the best baseline (PFedRec) are notable, with gains of 1.38%, 0.46%, 2.31%, and 1.09% in HR, respectively. For NDCG, APFR either outperforms or matches the best baseline, especially on larger or sparser datasets, highlighting its ability to recommend not only relevant but also well-ranked items.

Comparing the baseline methods, we find that centralized MF and FedMF achieve similar performance on most datasets, suggesting that the federated framework itself does not necessarily harm recommendation quality when user embeddings are kept private. FedNCF generally underperforms MF and FedMF, possibly due to optimization challenges or insufficient model capacity under privacy constraints. FedPerGNN yields significantly lower results across all datasets and metrics, likely due to the difficulty of effectively leveraging graph structures in highly sparse, federated settings.

PFedRec, the federated personalization baseline, outperforms all previous methods and achieves competitive performance, especially in NDCG on MovieLens-100K. However, APFR consistently surpasses PFedRec in almost all datasets, demonstrating the advantage of integrating attention mechanisms and improved personalization modules. Notably, the relative performance differences among methods are more pronounced on datasets such as Amazon Video and Lastfm-2K, where the modeling of complex user-item interactions and effective utilization of limited interaction

Table 2 Experimental parameter settings

Parameter	Setting
Client Sampling Rate	1.0
Total Training Rounds	100
Client Training Rounds	1
Learning Rate	0.1
Optimizer	SGD
Attention Mechanism Blocks	8

Table 3 Performance comparison of different models across datasets in terms of HR@10 and NDCG@10 (%)

Dataset	Metric	MF	FedMF	FedNCF	FedPerGNN	PFedRec	APFR
ML100K	HR	64.43	65.15	60.62	10.50	72.74	**74.12**
	NDCG	38.95	39.37	33.72	4.93	**44.83**	44.17
ML1M	HR	68.34	67.82	60.57	9.70	73.26	**73.72**
	NDCG	41.40	40.93	34.37	4.58	44.36	**44.55**
Lastfm	HR	82.25	81.74	81.53	10.35	81.81	**84.12**
	NDCG	70.98	69.29	61.02	4.96	71.77	**73.77**
Amazon	HR	46.62	59.67	57.80	10.63	59.46	**60.55**
	NDCG	29.91	38.55	37.19	4.90	37.73	**39.50**

Bold and underlined values represent the best results

data are especially critical. The superior results of APFR on these datasets indicate its robustness and enhanced modeling capacity for both dense and sparse recommendation scenarios.

Overall, these results demonstrate that APFR not only achieves state-of-the-art accuracy and ranking quality across a diverse set of real-world recommendation datasets but also consistently outperforms both traditional and advanced federated learning baselines. The observed improvements can be attributed to the effective incorporation of attention mechanisms and a more nuanced approach to user preference modeling within a privacy-preserving federated framework.

5.3 Ablation study and parameter sensitivity

We conducted ablation studies to verify the contribution of the attention mechanism and score function. As shown in Tables 4 and 5, the attention-based model achieves superior performance to the default model across most datasets and metrics for all five embedding sizes. Compared with APFR without the attention module, the HR of the APFR increases by 2.82% when the embedding size is set as 32. The same improvement also appears in the Amazon-Video. Under all embedding size values, HR and NDCG of APFR perform better than the model without attention. For MovieLens-100K and MovieLens-1M, attention performs better when the embedding size is low, which may indicate that the attention mechanism is better at extracting effective features in low-dimensional space. In addition, we conducted ablation studies to evaluate the contribution of personalized module in our model. As shown in Tables 4 and 5, the model without the score function, “APFR w/o Score”, consistently underperforms compared to the full APFR model. This highlights the importance of the personalized score function in improving the model’s performance. Removing the score function leads to a performance drop in HR and NDCG metrics across almost all datasets and embedding sizes, especially for MovieLens-100K and MovieLens-1M.

Table 4 Performance comparison between the APFR and its variants for different embedding sizes across datasets, evaluated using HR@10 (%)

Dataset	Model	4	8	16	32	64
MovieLens-100K	APFR	**67.44**	**73.06**	**73.38**	**74.12**	71.26
	APFR w/o Attention	66.80	72.74	73.17	72.74	70.94
	APFR w/o Score	67.33	71.36	73.17	71.79	**72.11**
MovieLens-1M	APFR	**62.94**	**69.70**	**73.17**	**73.72**	72.13
	APFR w/o Attention	62.54	69.45	72.76	73.26	**72.93**
	APFR w/o Score	61.10	68.92	72.58	73.21	71.52
Lastfm-2K	APFR	**84.25**	83.93	**84.31**	**84.12**	**84.06**
	APFR w/o Attention	83.56	**84.00**	82.68	81.81	82.68
	APFR w/o Score	83.68	82.87	83.00	82.81	82.43
Amazon-Video	APFR	**60.15**	**60.18**	**60.29**	**60.55**	59.79
	APFR w/o Attention	59.96	59.86	59.17	59.46	59.35
	APFR w/o Score	59.72	60.09	60.07	59.99	**59.86**

“APFR w/o Attention” refers to the APFR model without the attention mechanism. “APFR w/o Score” refers to the APFR model without the score function

Table 5 Performance comparison between the APFR, APFR w/o Attention, and APFR w/o Score models for different embedding sizes across datasets, evaluated using NDCG (%)

Dataset	Model	4	8	16	32	64
MovieLens-100K	APFR	**<u>38.15</u>**	**<u>42.25</u>**	**<u>43.91</u>**	44.17	**<u>43.99</u>**
	APFR w/o Attention	37.32	41.44	43.20	**<u>44.83</u>**	43.27
	APFR w/o Score	36.90	41.04	43.21	44.20	43.17
MovieLens-1M	APFR	**<u>35.28</u>**	**<u>39.97</u>**	**<u>43.32</u>**	**<u>44.55</u>**	43.93
	APFR w/o Attention	34.76	39.60	43.14	44.36	**<u>45.08</u>**
	APFR w/o Score	34.13	39.93	43.03	44.24	43.76
Lastfm-2K	APFR	**<u>74.22</u>**	**<u>74.00</u>**	**<u>73.81</u>**	**<u>73.77</u>**	**<u>73.98</u>**
	APFR w/o Attention	72.79	73.72	73.15	71.77	72.51
	APFR w/o Score	73.44	72.84	72.66	72.91	72.57
Amazon-Video	APFR	**<u>38.86</u>**	38.78	**<u>39.11</u>**	**<u>39.50</u>**	**<u>39.01</u>**
	APFR w/o Attention	38.05	37.54	37.40	37.73	37.41
	APFR w/o Score	38.76	**<u>38.91</u>**	38.90	38.65	38.65

Bold and underlined values represent the best results

For instance, in MovieLens-100K with embedding size 32, "APFR w/o Score" shows a 2.33% decrease in HR compared to the full model, which suggests that the score function plays a crucial role in capturing users' personalized preferences.

The Tables 4 and 5 demonstrate the influence of different embedding sizes toward the model integrating attention and without attention (marked as Default). The model performance generally increases as the embedding size increases. The improvement was significant, especially for small embedding sizes (*e.g.*, 4 and 8). However, the growth became gentle when large embedding sizes (*e.g.*, 32 and 64), and even some performance degradations were observed. *e.g.*, Attention model for MovieLens-100K, there were obvious improvements for HR and NDCG when embedding size from 4 to 32 but decreased significantly when embedding size 64 was used. This trend can also be observed in other datasets (*e.g.*, MovieLens-1M and Amazon-Video). In addition, the sensitivity across different datasets is slightly distinct. In the Lastfm-2K, even with large embedding dimensions, the model performance degradation was relatively small for HR, and even non-degraded on NDCG. This suggests that the value of embedding dimensions may also be related to the features of the dataset, such as the sparsity of user-item interactions or dataset size.

5.4 Privacy protection performance

To enhance user privacy protection, we integrated Local Differential Privacy (LDP) into APFR. The zero mean Laplace noise and white Gaussian noise were added into item embeddings before uploading them to sever:

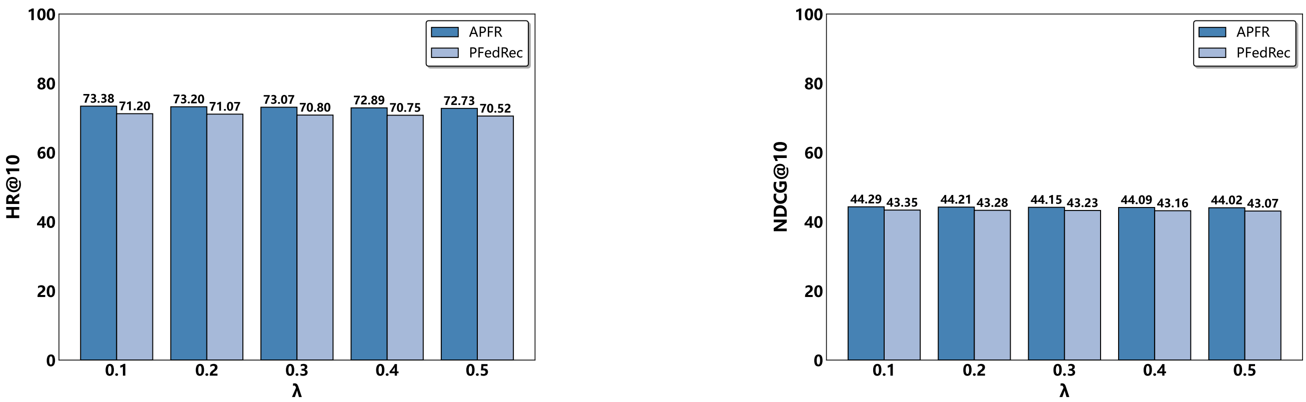
$$\begin{aligned} \theta_m &= \theta_m + \mathrm{Laplace}(0, \lambda) \\ \theta_m &= \theta_m + \mathrm{Gauss}(0, \lambda), \end{aligned} \tag{11}$$

where $\lambda = [0.1, 0.2, 0.3, 0.4, 0.5]$ refers to noise intensity; $\mathrm{Laplace}(0, \lambda)$ and $\mathrm{Gauss}(0, \lambda)$ are shown as (7) and (8).

The effects of different differential privacy mechanisms on recommendation performance are shown in Figs. 2, 3, 4, 5, 6, 7, 8 and 9. Overall, both APFR and PFedRec experience a slight decline in HR and NDCG as the noise intensity λ increases, which reflects the expected privacy-utility trade-off. Importantly, APFR consistently achieves better recommendation accuracy than PFedRec under all privacy settings, demonstrating its superior robustness to privacy-preserving noise across all benchmark datasets.

For the Laplace noise mechanism, as illustrated in Figs. 2 to 5, APFR maintains a clear performance advantage over PFedRec at all levels of noise intensity on every dataset. The HR and NDCG scores of APFR remain stably higher than those of PFedRec, indicating that the proposed attention-based modeling and personalized Score functions make APFR less sensitive to the perturbations introduced by Laplace noise. This robustness is observed regardless of dataset scale or sparsity, highlighting APFR's adaptability to a variety of real-world recommendation scenarios.

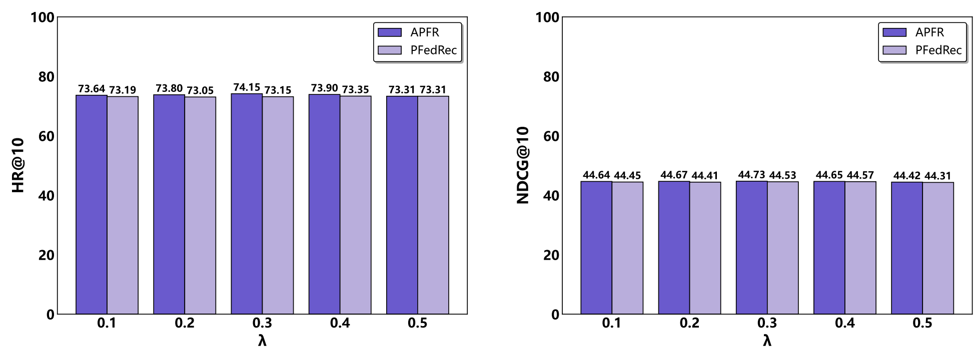
Under the Gaussian noise mechanism, as depicted in Figs. 6 to 9, APFR continues to outperform PFedRec consistently. The performance trends are similar to those observed with Laplace noise, further confirming that APFR is generally insensitive to the specific type of differential privacy noise injected into the federated learning process. These results suggest that the attention mechanism within APFR effectively preserves key user-item interaction information even when model updates are heavily perturbed, leading to more reliable and robust recommendation outcomes. The results across both privacy mechanisms demonstrate that APFR provides strong privacy protection with minimal loss in recommendation quality. Its stability and robustness under



(a) HR@10 (b) NDCG@10

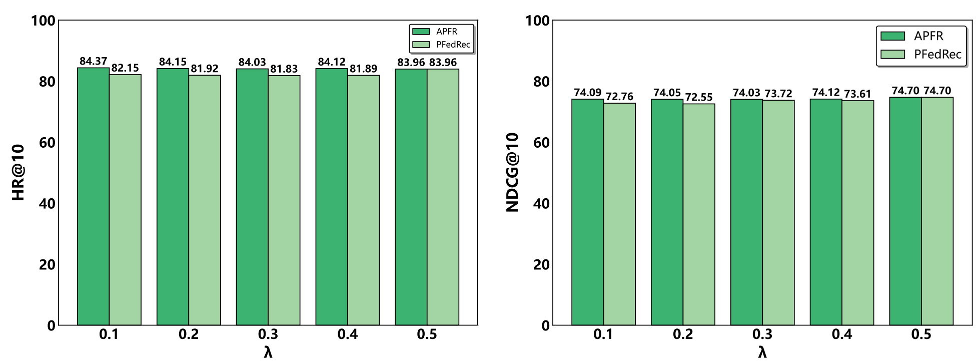
Fig. 2 Comparison results for the Movielens-100K dataset under Laplace noise

Fig. 3 Comparison results for the Movielens-1M dataset under Laplace noise



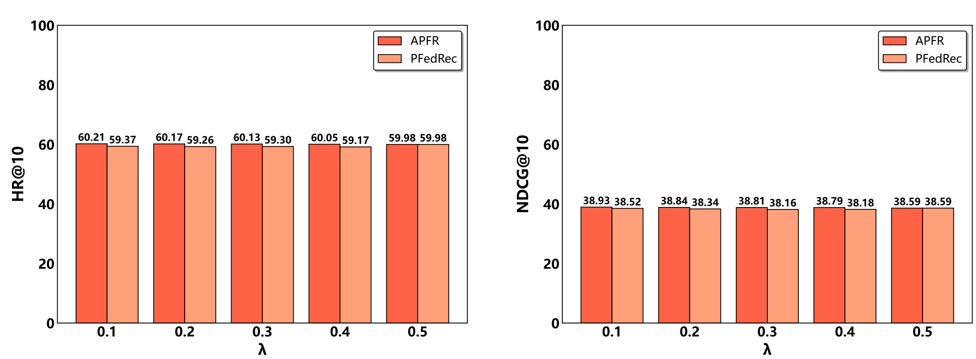
(a) HR@10 (b) NDCG@10

Fig. 4 Comparison results for the Lastfm-2K dataset under Laplace noise



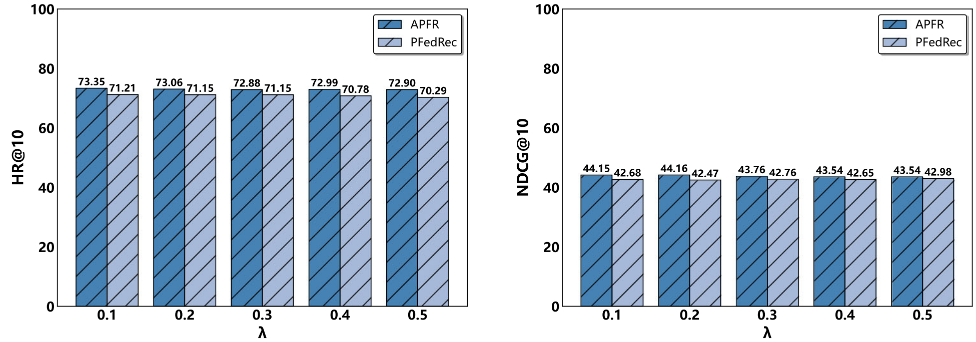
(a) HR@10 (b) NDCG@10

Fig. 5 Comparison results for the Amazon-Video dataset under Laplace noise



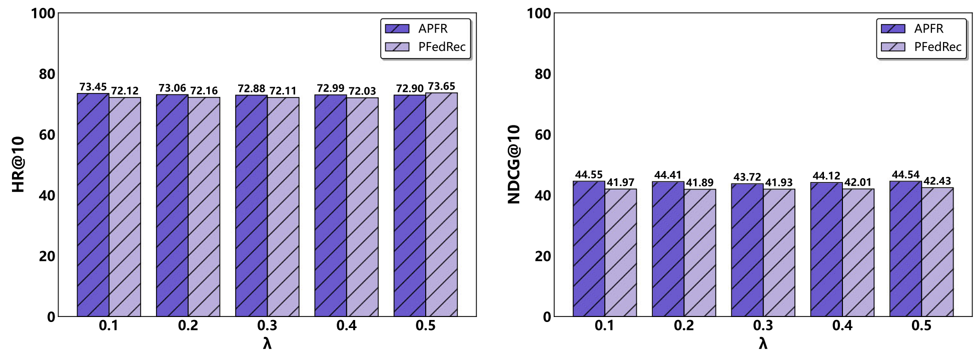
(a) HR@10 (b) NDCG@10

Fig. 6 Comparison results for the Movielens-100K dataset under Gaussian noise



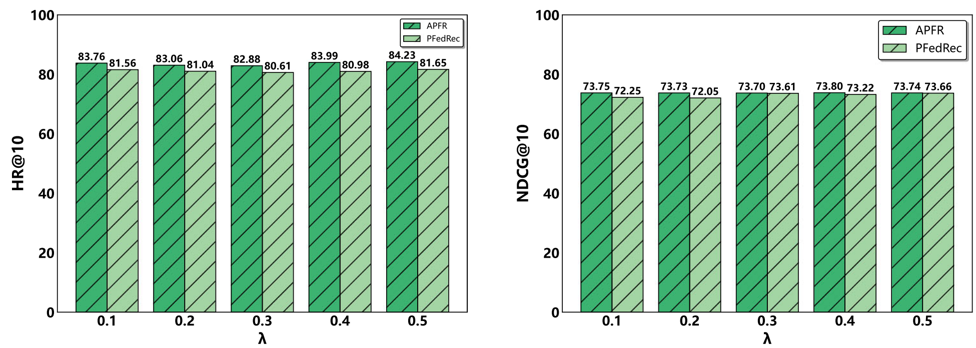
(a) HR@10 (b) NDCG@10

Fig. 7 Comparison results for the Movielens-1M dataset under Gaussian noise



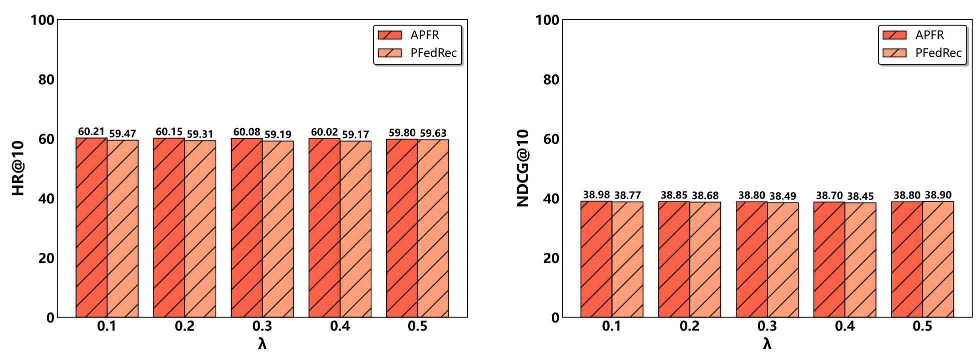
(a) HR@10 (b) NDCG@10

Fig. 8 Comparison results for the Lastfm-2K dataset under Gaussian noise



(a) HR@10 (b) NDCG@10

Fig. 9 Comparison results for the Amazon-Video dataset under Gaussian noise



(a) HR@10 (b) NDCG@10

different noise types and intensity levels make it a practical and reliable solution for privacy-preserving federated recommendation systems.

6 Conclusion

In this work, we proposed the APFR framework, which integrates attention mechanisms into federated learning to enhance the accuracy and personalization of recommendations. Unlike existing federated recommendation models, APFR employs both personalized score functions and user embeddings, and leverages self-attention and multi-head attention to capture complex item relationships and contextual dependencies. Extensive experiments on four benchmark datasets—MovieLens-100K, MovieLens-1M, Lastfm-2K, and Amazon-Video—demonstrate that APFR consistently outperforms state-of-the-art centralized and federated baselines in both accuracy and robustness to noise introduced by differential privacy. Even under varying levels of Laplace or Gaussian noise, APFR maintains a substantial performance advantage over competing models, highlighting its practical value for privacy-preserving personalized recommendation in distributed environments.

Despite these promising results, several limitations remain to be addressed. First, while APFR effectively balances personalization and privacy, the current framework does not explicitly address challenges related to model scalability and computational efficiency on edge devices, especially in scenarios with a rapidly expanding item or user base. Besides, although the attention mechanism improves accuracy, it increases model complexity and may reduce the interpretability of recommendations for end users. Moving forward, it will be valuable to develop lightweight and scalable model architectures suitable for deployment in resource-constrained environments, as well as to design more transparent and interpretable attention mechanisms that can enhance user trust. These efforts will help pave the way for the broader and more responsible application of federated learning in real-world personalized recommendation systems.

Acknowledgements This research was supported in part by the National Natural Science Foundation of China [No. 62107022], and in part by the Startup Fund of Jimei University [No.ZQ2024014].

Author Contributions L. Zheng: Conceptualization, Methodology, Software. H. Chen: Data curation, Writing—original draft. Y. Lin: Project administration, Funding acquisition, Writing—original draft. L. Li: Formal analysis, Writing—review and editing. Y. Yu: Visualization. H. Chen: Validation. K. Li: Investigation. R. Hu: Supervision, Writing—review and editing. S. Xie: Resources.

Data availability statement The datasets used and/or analyzed during the current study are available at https://github.com/lyg2618/APFR/tree/main/data.

Declarations

Competing interests The authors declare that they have no competing interests.

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